



Partitioned Convolutional Spiking Neural Network for Tactile Object Recognition

Jun Sung Go¹ · Jong Tae Kim¹

Received: 25 August 2025 / Accepted: 10 December 2025 / Published online: 6 January 2026
© The Author(s) 2025

Abstract

The human tactile system, which occupies the largest sensory area, has spurred active research on artificial tactile systems for applications in robotics, virtual reality, and e-skin technologies. Applying deep learning models to analyze large-scale, continuous temporal data and detect patterns in such systems is a well-established methodology. However, mimicking biological models for processing large amounts of sequential data is a potentially more economical approach. These biologically inspired models are expected to be more efficient than traditional deep learning models that rely on central processing units (CPUs) or graphics processing units (GPUs). In this study, we implemented an artificial tactile system using a partitioned spiking neural network (SNN). This approach was applied to classify objects detected using an artificial glove equipped with tactile sensors. Our method achieved higher accuracy than the existing SNN architecture. Furthermore, the reduced number of sensor readout cycles and computational requirements demonstrate the potential for faster results than conventional methods.

Keywords Tactile system · Convolutional spiking neural network (CSNN) · Spike-timing-dependency plasticity (STDP) · Gabor filter · Scalable tactile glove (STAG)

1 Introduction

Touch is the most widespread mechanoreceptor system in the human body. Thousands of nerve fibers detect pressure on the skin and sense the roughness of objects by moving hair or deforming tissues such as skin or mucous membranes [1]. These fibers help humans feel the size and shape of an object [2]. With different types of touch receptors, humans learn to interact with and understand the world and handle objects with high sensitivity.

✉ Jong Tae Kim
jtkim@skku.edu

Jun Sung Go
myun243@skku.edu

¹ Department of Electrical and Computer Engineering, Sungkyunkwan University, Seoul 16419, Republic of Korea

Interest in touch has led to the development of artificial tactile systems for robots, such as robotic arms, which attempt to copy human behavior. Research has also shown that these systems can be used in wearable devices for virtual and augmented reality, as in the metaverses [3–5]. Researchers are working on creating artificial skin using nanomaterials such as piezoelectric sensors, which can sense pressure and vibrations and turn them into signals for real touch [6]. In addition, soft e-skin can be added to robots to help them handle and recognize objects similarly to humans [7–10].

Interpreting high-dimensional data from tactile sensors requires the discovery of patterns in input signals. Artificial neural networks (ANNs) are used to recognize sensory information by modeling human biological characteristics. A structure composed of numerous parallel artificial neurons and synapses processes data and solves complex practical problems [11, 12].

One of the main challenges in tactile systems is the continuous influx of extensive time-sequenced tactile data. In humans, over 45,000 mechanoreceptors are partitioned throughout the body, with more than 18,000 located on the hands [13]. Because tactile signals are continuously transmitted through the system, significant energy consumption and response time delays occur owing to limited bandwidth. When processing large amounts of continuous time-series data using ANNs with high-performance GPUs, substantial power is required for sensory cognitive operations [14]. Moreover, with a tactile sensor that uses the conventional time-division multiplexing method, the number of cycles required to read the entire dataset increases as the data size increases. As a result, a single global network is forced to compute and scan regions with no informative content, and readout latency and data movement accumulate as the frame scale grows.

Biologically, the human tactile system performs peripheral, receptive-field-level preprocessing that forwards only key timing cues to higher areas, thereby compressing information locally and reducing both energy use and the computational load on downstream processors. Reducing the load during the high-level interpretation of tactile information requires the preprocessing of sensory signals for individual parts using sensors and processing units (PUs), instead of processing the entire tactile input simultaneously [15].

A spiking neural network (SNN), also known as a third-generation ANN, processes information by transmitting electrical signals called spikes, similar to the biological nervous system. It transmits data using the relative timing of binary spikes and learns through the time difference between the input and output spike generation using unsupervised learning methods, such as spike-timing-dependent plasticity (STDP) [16, 17]. This approach solves the power consumption issues in traditional ANNs that use floating-point multiplication for synaptic weights and activation functions [16, 18]. In addition, spike-based representations reduce the amount of data carried during information transmission. Recent studies have demonstrated that SNNs can achieve low power consumption while preserving the accuracy of conventional ANNs owing to advancements in materials such as memristors [19–22].

In this study, we propose a tactile system using a partitioned convolutional spiking neural network (CSNN) architecture. Here, “partitioned” means that the input frame is divided into anatomically meaningful regions; each region is processed by a local CSNN that extracts features, and a central unit concatenates only these regional features to make the final decision. This network applies a CSNN composed of temporal coding, convolutional, and pooling layers to each region of the hand to extract region-specific feature information. The acquired information was merged into a concatenation layer, and the detected object was classified in the final classification layer. The filter sizes in the convolutional layers were adjusted to match the different sizes of the palm and finger regions, thereby improving the feature extraction efficiency and recognition accuracy. Unlike previous methods that recognize hand

information in a rectangular format, this approach omits computations for empty spaces. This segmented structure reduces the number of sensor data reading cycles via partition-wise scanning compared with methods that read all the sensor data simultaneously. Furthermore, by preprocessing the data with a partitioned neural network structure, the computational load on the central processing unit is reduced, allowing for faster object recognition, and the reduction in data movement alleviates the compute–memory bottleneck.

This work introduces a partitioned CSNN that aligns the model with the spatial sparsity and temporal redundancy of tactile inputs. By eliminating empty-space computations and scanning each anatomical partition independently, the design shortens sensor readout cycles and reduces data movement. The network further leverages spike-latency representations that preserve discriminative contact transients, yielding accuracy comparable to conventional ANNs while requiring fewer operations.

The rest of the paper is structured as follows: Sect. 2 addresses related work on tactile perception through artificial intelligence. Section 3 details the structural model and Sect. 4 explains the architecture, basic components, and principles of a partitioned CSNN. Section 5 presents the experimental environment for tactile perception and reports the experimental process and results. Finally, Sect. 6 presents conclusions and discusses future directions.

2 Related Work

Deep learning has been widely applied to numerous recognition systems based on visual and auditory information. Recently, it has also been applied to recognize objects by touch using pressure data from sensors embedded in artificial skin. This section presents an overview of previous research on tactile sensing using deep learning neural networks.

In [23], indicators, including activity, mobility, and complexity, were determined using vibration and contact signals captured by multimodal tactile sensors. These findings were subsequently applied to a least-squares support vector machine to perform object recognition experiments based on surface texture detection. In [24], a sensor mimicking the human fingertip was used to detect fine vibrations, forces, and temperatures. The detected data were processed using a network combining a support vector machine (SVM) and an artificial neural network (ANN) to analyze thermal conductivity and surface texture for material classification. Consequently, the classification accuracy is approximately 7% higher than that achieved using direct human touch. In [25], a planar flexible sensor and deep neural network enabled the ultraprecise detection of pressure point numbers, shapes, sizes, and locations with few electrodes and simplified signal processing.

Convolutional neural networks (CNNs) are neural networks that effectively extract features, making them useful for processing visual information and classification [26, 27]. Recently, numerous studies have applied CNNs to tactile systems, based on the similarity between visual and tactile information, to extract various object attributes. In [28], the roughness of an object was measured by applying a CNN to the continuously acquired time data detected by a pen-type sensor combined with a pressure sensor and a 6-axis accelerometer while moving it over the object's surface. The study showed that the surface roughness of objects with a size of 1.6 could be recognized. In another study [29], a CNN was used to recognize 11 attributes of clothing, including thickness, pilling, and softness, using GelSight tactile sensors.

Another study was conducted using SNN for tactile perception. In [30], object shapes were recognized using information detected by 42 tactile sensor channels through an SNN

composed of Izhikevich neurons and excitatory and inhibitory synapses. Hebbian learning using a pair-based STDP algorithm achieved 100% accuracy in recognizing object shapes. In [31], the authors recognized the edge direction, which is one of the main functions of the fingertip, by modeling the three stages of the tactile pathway from the skin mechanoreceptors to the cortex using a spike-based neural network. [32] and [33] demonstrated object classification using a Spiking Graph Neural Network by constructing a tactile graph from tactile data sensed by NeuTouch, an event-based tactile sensor with 39 taxels.

Previous studies have demonstrated high-accuracy tactile perception and classification systems using SNN architectures. However, these studies were limited in their ability to model complex and large-scale tactile systems such as those resembling the human hand because of the use of small-scale data focused on fingertip-level interactions. As a result, partition processing was not implemented, and the entire dataset was received and processed simultaneously by a central device. This paper presents an approach that differs from previous research in that it receives only a reduced form of the original data from the main device, enabling the rapid and energy-efficient processing of large-scale tactile data.

3 Structural Model of Partitioned CSNN

Figure 1 shows the overall structure and operation of the partitioned CSNNs used in this study. The pressure data detected in each region of the hand were segmented and used as inputs for separate CSNNs corresponding to each part. Figure 2 shows the components of a single-layer CSNN. First, the pressure data measured by the sensor were converted into spike trains with temporal information through the input encoding layer. These spikes are then input into the convolution layer, where complex features are derived based on common local characteristics. The filter size in the convolution layer is adjusted based on the size of the input data. The pooling layer compresses the pressure data by applying a maximum operation and propagates the fastest spike received from the neighboring neurons. The feature outputs extracted from each CSNN are combined into a single piece of information using a concatenation layer. Finally, the combined feature information is applied to a classifier to identify the object detected by the hand.

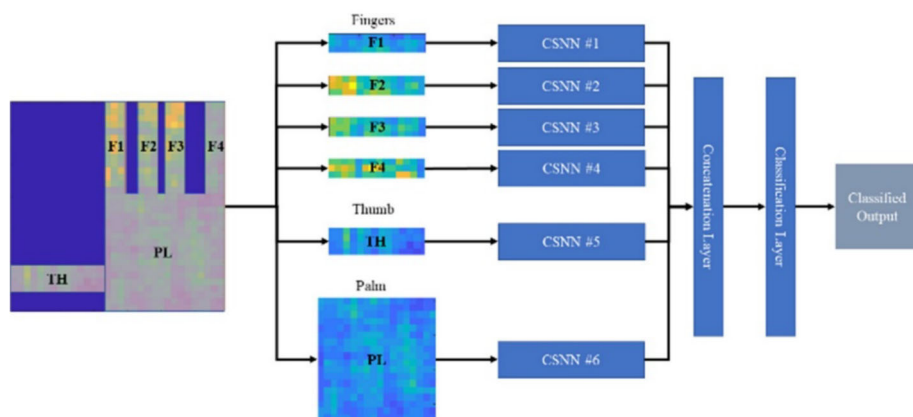


Fig. 1 Overall structure of the partitioned CSNN

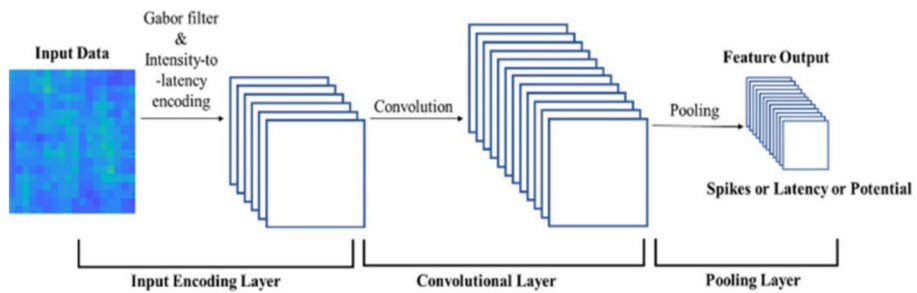


Fig. 2 Architecture of CSNN for feature extraction

4 Architecture of CSNNs

4.1 Input Encoding

A preprocessing step is required to convert the images into spike information before they can be input into an SNN. In this study, as shown in Fig. 3, the input tactile pressure signal is converted into time-dependent spike information in two steps. The Gabor filter, a type of bandpass filter, is well-known for its ability to extract various local structures from images or videos. Comprising a Gaussian function and a modulated complex sinusoidal component, the Gabor filter offers excellent spatial frequency localization performance. According to previous studies [34], Gabor filter has been shown to effectively emulate the human visual system, suggesting that it can also be efficiently utilized in modeling tactile systems.

The edge features extracted by the Gabor filter must be converted into spike data. Spike data is typically categorized into rate coding and temporal coding. Rate coding, which converts input values into spike arrays with frequencies ranging from 0 to 63.75 Hz through the Poisson transform [35], conveys information through firing intensity. In this case, the number of synaptic-spike operations increases by the number of input spikes during data transmission.

Temporal coding conveys sensory input information through specific spike times of a single neuron or a group of neurons. This approach reduces the number of synaptic-spike convolution operations, thereby minimizing the computational load on the network.

In this paper, we employ a temporal coding scheme in which, as illustrated in Fig. 3, pressure intensity is represented by a single spike within a prescribed time window. Under the intensity-to-latency encoding, higher input amplitudes are mapped to a single spike that occurs earlier on a discrete-time axis with sampling period.

4.2 Convolutional Layers

The convolutional layer applies multiple convolution filters to the data from adjacent pixels in an image, outputting a feature map that captures the spatial information of the image. This enables the extraction of common features from different parts of an image, thereby enhancing its interpretability.

Among the types of artificial neurons used in SNNs, the non-leaky integrate-and-fire neuron [36] is the simplest. The membrane potential of this neuron increases based on the product of the synaptic weights and spike output from the presynaptic neurons (Fig. 4a). The

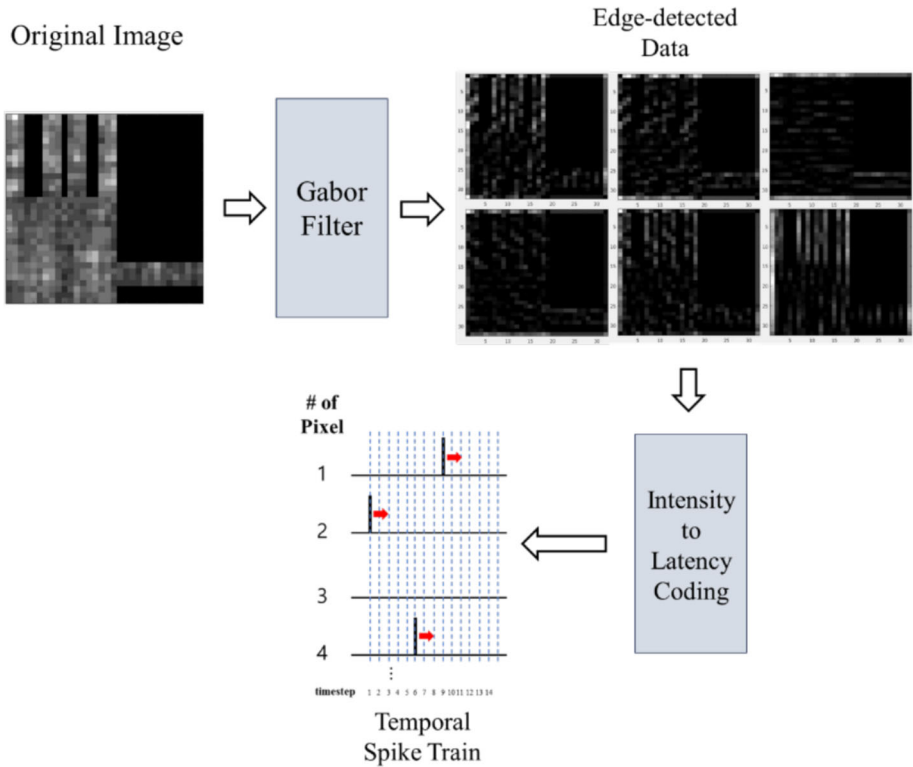


Fig. 3 Flow of the input encoding system. The input image is first converted into edge-detected data through a Gabor filter, and then the converted data is converted into a spike train that emits at most one spike per cycle through intensity-to-latency coding

membrane potential of the output neuron over time is expressed as

$$V_O(t) = V_O(t-1) + \sum_i w_{i,o} S_i(t-1) \quad (1)$$

where $V_O(t)$ is the membrane potential of the output neuron at time step t . $w_{i,o}$ is the synaptic weight between the input and output neurons, and $S_i(t-1)$ is the spike output from the input neuron. If $V_O(t)$ crosses the membrane threshold, the output neuron produces a spike and resets the membrane potential (Fig. 4b).

$$V_O(t) = 0 \text{ and } S_O(t) = 1, \text{ if } V_O(t) \geq V_{th} \quad (2)$$

In the convolutional layer, lateral inhibition causes a neuron to fire and suppresses nearby neurons, resetting their membrane potential to zero until the next image is input. The fired neuron remains inactive until the next input.

4.3 Spike-Time Dependency Plasticity

One of the characteristics of SNNs is their use of unsupervised learning through STDP. The synaptic weight increases if the presynaptic spike contributes to the firing of the postsynaptic

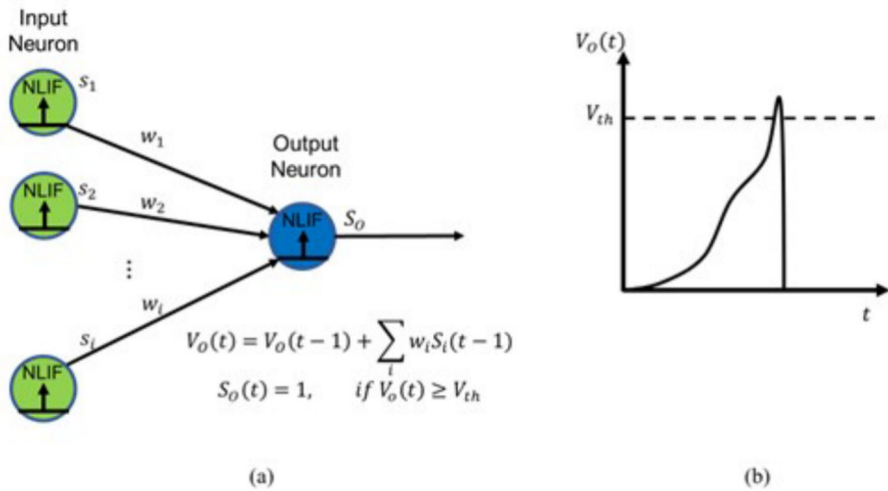


Fig. 4 Characteristics of spiking neurons. **a** Membrane voltage of output neuron is determined by the spikes that occur on the input neuron and the weight of the synapse between the input neurons and the output neuron. **b** When the membrane voltage exceeds the threshold, output neuron emits the spike and resets its membrane to 0

neuron based on the time difference between them and decreases otherwise. In this paper, the weight between the postsynaptic neuron j and the presynaptic neuron i is updated using a simplified version of STDP [37],

$$\Delta w_{i,j} = \begin{cases} a^+ w_{i,j} (1 - w_{i,j}), & \text{if } t_j - t_i \leq 0, \\ a^- w_{i,j} (1 - w_{i,j}), & \text{if } t_j - t_i > 0 \end{cases} \quad (3)$$

where t_i and t_j represent the firing times of the neurons, and a^+ and a^- denote the learning rates for Long-Term Potentiation (LTP) and Long-Term Depression (LTD), respectively. The positive coefficient, a^+ , regulates the magnitude of LTP, while the negative coefficient, a^- , controls the magnitude of LTD.

In the convolutional layer, the kernels detect the same feature across all regions but compete to identify different features. The winner-take-all algorithm is used to designate the first-firing neuron as the winner within the feature map, where STDP is applied only to that neuron and is limited to all neurons within the same feature map. Simultaneously, the STDP in adjacent neurons in other feature maps near the location where STDP occurred is also restricted, enabling differentiated learning for the kernels in relation to the features.

4.4 Pooling, Concatenation Layer

The primary purpose of the pooling layer is to reduce the size of the feature maps extracted from the convolutional layer to enhance the data features. This layer compresses visual information and decreases the size of the subsequent layers, thereby increasing the computational efficiency of the neural network. The neurons in the pooling layer perform the maximum operation over a window in the corresponding neuron map of the previous layer. The synaptic weights and threshold values of the neurons in the pooling layer are set to one to ensure that

the first input spike from the neurons in the convolutional layer becomes the output spike in the pooling layer.

The concatenation layer combines the feature information from each region extracted from the previously partitioned CSNN into a single dataset. This combined information includes feature data for the entire hand, which is then input into the final layer, the classification layer, to classify the output class of the detected object.

4.5 Classifier Layer

Three classifier methods were tested in the experiment: k-nearest neighbor (KNN), support vector machine (SVM), and extremely randomized trees (Extra-Trees).

KNN classifies an input data point by finding the k-nearest neighbors in the training dataset and assigning the majority class label to those neighbors. Distance metrics, such as Euclidean or Manhattan distances, determine the closeness of neighbors. For objects with similar shapes, sizes, or weights, KNN is well-suited for grouping or distinguishing objects based on their proximity in the data space.

SVMs aim to find an optimal hyperplane that maximizes the margin between different classes. This property makes them highly robust in classification tasks, especially in cases where class separation is not trivial. For datasets that are not linearly separable, SVM employs the kernel trick, mapping data into higher-dimensional feature spaces to capture nonlinear relationships.

Extra-Trees are ensemble models that train numerous decision trees and aggregate their results to produce predictions. Extra-Trees train decision trees using the entire dataset and randomly select split values for features to create sub-nodes, ensuring sufficient differences between individual trees. This process typically results in faster speeds and better performance than the previous Random Forest classifier method.

5 Implementation & Results

5.1 Datasets

The Scalable Tactile Glove (STAG) dataset [38] is a collection of pressure data captured while touching or grasping objects. A total of 548 pressure sensors, distributed as listed in Table 1, were attached across the entire surface of the gloves. The sensors are piezo-resistive and made of soft materials, with each sensor capable of detecting vertical forces ranging from 30 mN to 0.5 N across 150 discrete levels. The pressure detected by the sensors was recorded as the normal force for each sensor using a readout circuit at an approximate frequency of 7.3 Hz (Fig. 5a). The recorded pressure data were stored in the form of images corresponding to the hand regions, as illustrated in Fig. 5b.

Table 1 Distribution of sensors on the scalable tactile glove

Sensor	Number of sets	Resolution (pixels)	Total pixel
Palm	1	18 × 18	324
Thumb	1	4 × 14	56
Finger	4	3 × 14	168

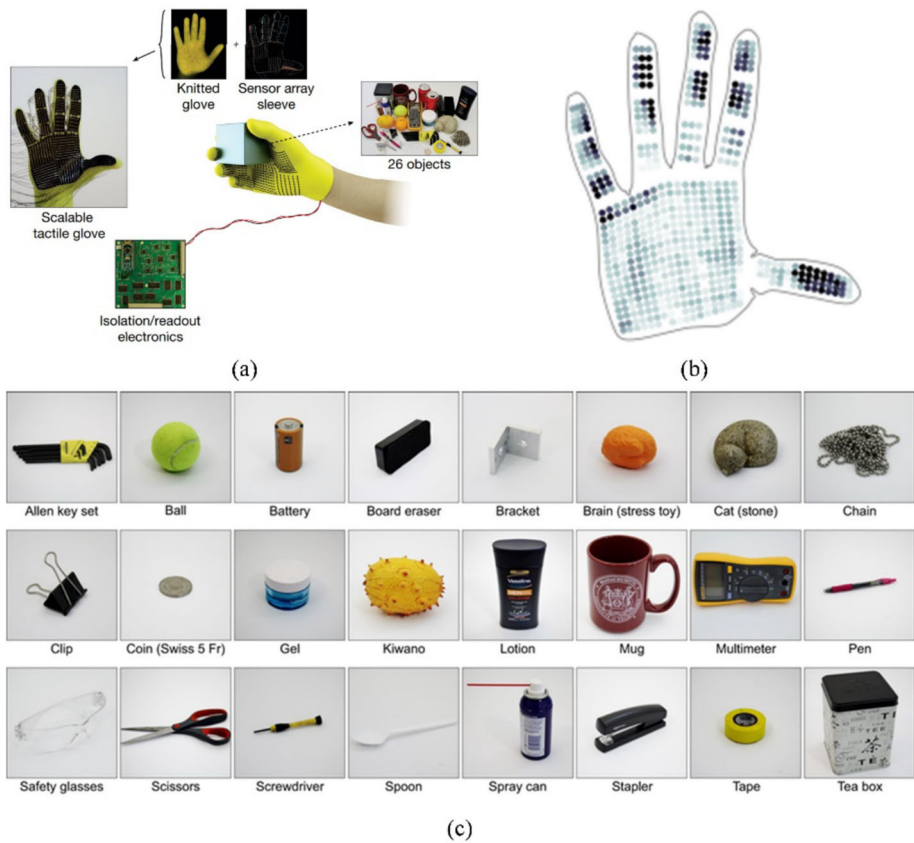


Fig. 5 **a** The structural characteristics of STAG. The STAG features a 548-element sensor array integrated into a custom knit glove. When manipulating objects, the normal force measured by the STAG is acquired through the readout circuit. **b** Tactile pressure map during single-hand manipulation of tea box. **c** Objects of STAG dataset. [38]

In addition to utilizing individual image frames, object classification was conducted using continuous video sequences of two to eight frames that captured the dynamic grasping process. This method allows for the extraction of high-level features and relationships from high-dimensional tactile data, leveraging advanced machine learning techniques to gain deeper insights into tactile feedback related to human perception.

The dataset was recorded during interactions with a set of 26 objects, as shown in Fig. 5c. From the entire dataset, 88,269 valid data points were retained after excluding instances with abnormally high values caused by electrode discontinuities resulting from contact with sharp metal objects. Of these, 36,531 data points, comprising 1,353 images per class, were used for training, whereas 16,119 data points, comprising 597 images per class, were used for testing.

6 Results

We performed experiments using the STAG dataset with partitioned CSNN architecture, as shown in Fig. 1. To determine the appropriate number of Gabor filters required in the

input-encoding stage (Fig. 3), we applied only a Gabor filter bank to the tactile-glove dataset and measured the classification accuracy of the resulting features without employing any additional network. To uniformly capture edge orientation, the filter orientations were evenly spaced according to the number of filters. Varying the number of filters from 2 to 8, we found that the six-orientation configuration achieved the highest accuracy (Table 2), indicating that a bank of six filters most effectively extracts the edge-like characteristics present in tactile data.

The convolutional layer sizes and the threshold of spiking neurons were customized for different regions, with separate configurations applied to the finger and palm data as in Table 3. Specifically, for finger data, where the smallest feature (finger width of 3 pixels) needed to be captured, 16 convolutional filters of size 3×3 were used. For the convolutional filter size corresponding to the palm, we repeated the experiment by changing the size from 3 to 9 to determine the optimal filter size N that yielded the highest accuracy. After each convolutional layer, the data were downsampled using a 2×2 pooling window. The test was performed using three classification methods: KNN, SVM, and Extra-Trees.

Experiments were conducted not only on a single frame, but also using multiple input frames ranging from 2 to 8, mimicking human-like tactile perception processes. As shown in Fig. 6, multiple partitioned CSNNs trained on a single frame were utilized, and their outputs were fused in the concatenation layer based on the number of input frames before being classified by a classifier. To evaluate the performance of the CSNN architecture more accurately, the object-detection accuracy was compared using three types of feature outputs—potential of output neuron, output spike occurrence, and spike latency as shown in Fig. 2.

As shown in Table 4, when the recognition experiment was repeated while changing the size N of the filter applied to the palm, the accuracy was highest when the filter size was 5. Although the Table 4 shows the results when latency was the output, it was confirmed that the accuracy was also highest when the filter size was 5 when spike or potential was the output.

Figure 7a presents the classification accuracy achieved by each of the three classifiers. KNN exhibited the highest performance among all methods when a single frame was used as input. This can be explained by the fact that KNN works well for grouping or distinguishing objects with similar geometric and physical properties, such as shape, size, or weight. However, when more frames were used, the accuracy declined significantly. This phenomenon occurs because, as the number of dimensions in the data increases, the data volume expands, causing the data to become sparse. Consequently, KNN struggles to find meaningful neighbors, and the distances between points become uniform, making differentiation difficult. In

Table 2 Classification accuracy using only Gabor filters on the tactile glove dataset

# of Filter	2	3	4	5	6	7	8
Accuracy (%)	23.49	27.48	27.19	27.14	27.81	27.56	27.7

Table 3 Parameters for convolutional layers for feature extraction

Sensor	Filter size (depth, width, height)	Threshold
Palm	(32, N, N)	2
Thumb	(16, 3, 3)	0.5
Finger	(16, 3, 3)	0.5

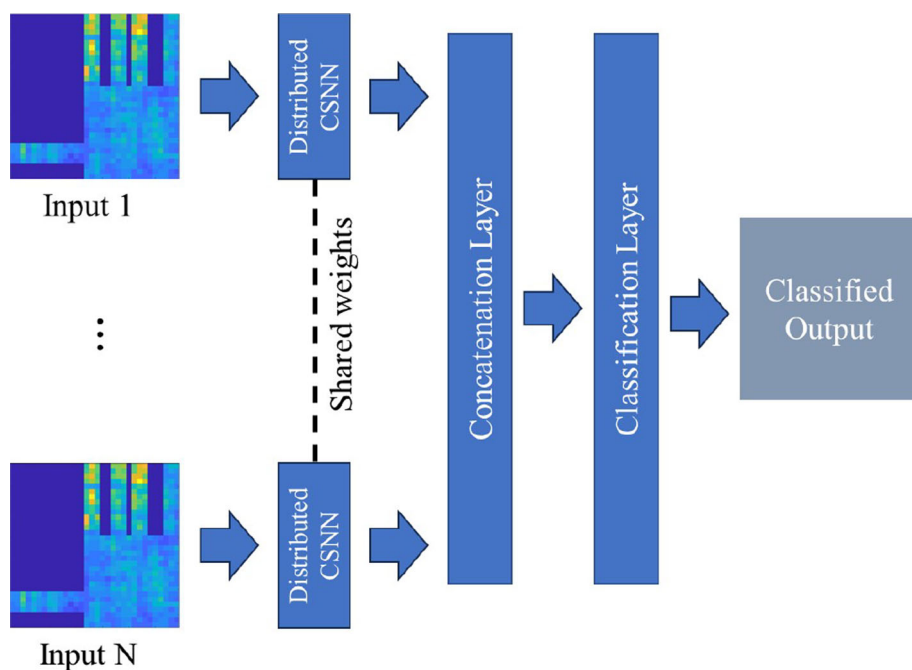


Fig. 6 Overall architecture used for identifying objects from tactile information takes as input N arrays of tactile data

Table 4 Classification accuracy varying size of convolutional filter at palm when having latency output

Size of Filter	3	5	7	9
Accuracy (%)	30.51	32.19	31.77	32.12

addition, this leads to overfitting, in which the model learns noise from the training data rather than from the actual patterns.

In contrast, SVM showed the lowest performance among all methods when using a single frame. However, as the number of input frames increased, its accuracy improved more rapidly than other methods, eventually reaching a level comparable to ExtraTrees for datasets with seven or more frames. Conversely, ExtraTrees achieved the highest accuracy for datasets containing two to six frames, but the accuracy was saturated when the number of frames reached seven or more. This is due to ExtraTrees' ability to capture essential patterns from a small set of frames, whereas SVM leverages higher dimensional input to refine its decision boundaries as more frames are added.

In all three cases, spike latency as the output consistently achieved the highest classification accuracy. This superiority stems from its ability to encode both stimulus intensity and temporal dynamics, thereby providing a richer and more discriminative representation. In contrast, spike occurrence conveys only binary information, while the final membrane potential continues to accumulate even after spike generation, leading to the loss of fine-grained temporal features.

Figure 8 presents the confusion matrices for the three outputs, illustrating the classifica-

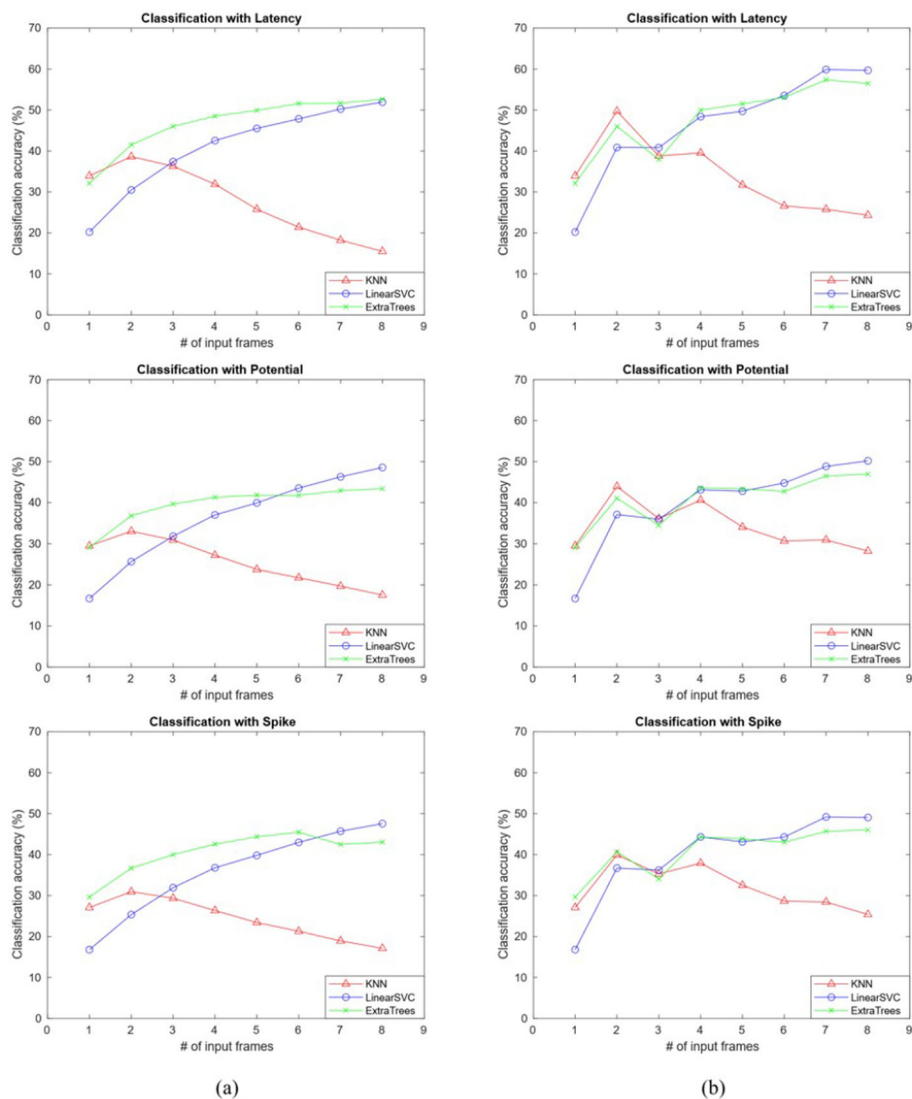


Fig. 7 **a** Object identification accuracy with randomly selected N frames using three classifiers and **b** with clustered N frames using three classifiers

tion performance for each category. The results indicate that while the performance variance across the three outputs is minimal, there is a significant discrepancy in accuracy among the categories themselves. Analysis of the matrices reveals a distinct trend: a higher incidence of inter-class confusion occurs among objects with lower physical weights. Specifically, lightweight items such as safety glasses, the board eraser, and the coin are frequently misclassified. In contrast, heavier objects possessing more distinct signatures—notably the tea box, cat (stone), and mug—demonstrate high classification accuracy.

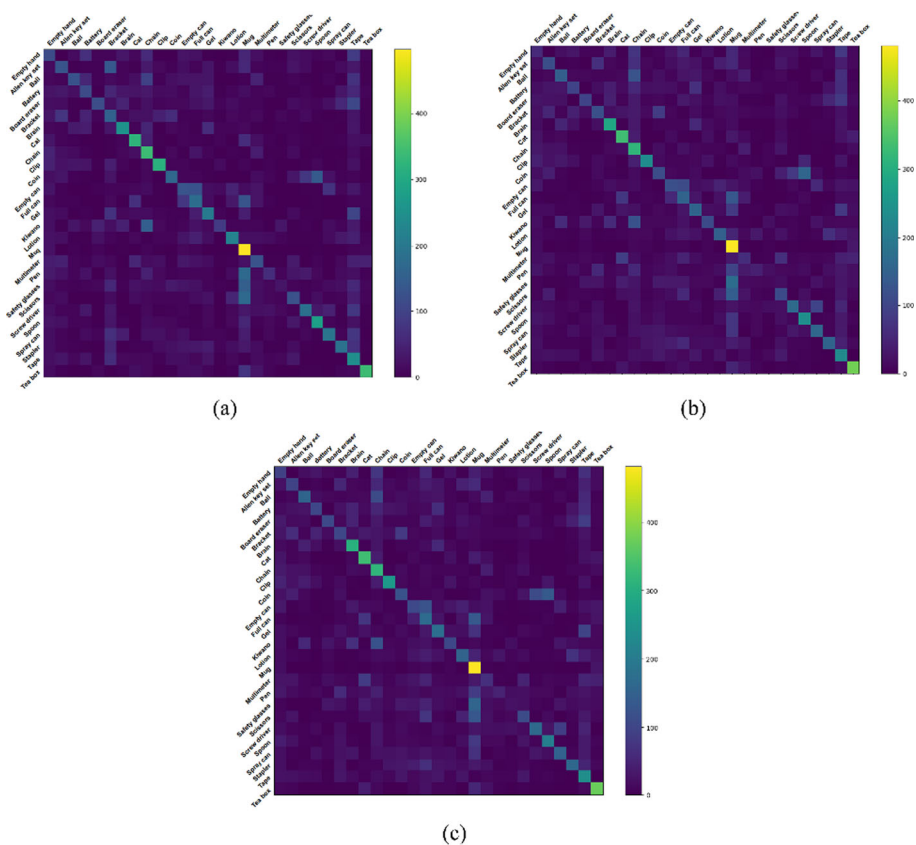


Fig. 8 Confusion matrix while using **a** latency, **b** potential and **c** spike

Humans can easily associate similar grasp movements with motor movements, and object identification tends to be more effective in selecting the most distinct and informative grasp actions. Based on this principle, the STAG dataset identifies the most diverse set of N frames from the input recording using k -means clustering instead of selecting N random frames.

When objects were classified using this clustered dataset, as shown in Fig. 7b, a significant improvement in overall accuracy was observed compared with using randomly selected input frames. Interestingly, the performance exhibited a nonlinear trend: classification accuracy steadily improved when using one to two frames, dropped sharply at three frames, and then increased again from four frames onward. This pattern may be due to an inherent limitation of the k -means clustering-based frame selection method.

K -means selects frames by maximizing the distance between clusters to ensure input diversity. While the initial frames tend to capture the most representative tactile features, the third frame is often selected for its distinctiveness, which may result in choosing frames that are less representative or contain noise. These outlier frames can temporarily disrupt the classifier's decision process, leading to decreased performance. As additional frames are incorporated (four or more), the model receives enough meaningful information to relearn stable patterns, allowing the accuracy to recover.

Table 5 compares the recognition accuracy of the proposed CSNN model with those of

Table 5 Recognition accuracies if the proposed partitioned CSNN and some other methods over the STAG dataset

Method	Accuracy using 1 Frame (%)	Accuracy using 8 Frames (%)
ResNet-18 (CNN) [38]	36.67	67.89
Spiking DCNN [39]	26.63	49.58
2-Layered (64-128) CSNN	28.89	51.21
Partitioned CSNN (latency-based) + KNN	33.95	24.34
Partitioned CSNN (latency-based) + SVM	20.22	59.9
Partitioned CSNN (latency-based) + Extra-Trees	32.19	57.4

the other models when applied to STAG datasets. In a previous study [38], a ResNet-18-based CNN architecture was used to achieve an accuracy of 36.67% for a single frame and 67.89% for 8 frames. This discrepancy results in an approximately 3% accuracy difference per single frame compared to the latency-based partitioned CSNN utilizing E-Trees, and an 8% accuracy difference over 8 frames compared to the SVM approach. This discrepancy arises from the fact that CNN-based ResNet-18 employs floating-point operations, allowing for richer information representation, and relies on a significantly more complex multi-layer architecture that demands substantially higher computational resources.

Compared to previous Spiking DCNN studies, which equals to non-partitioned SCNN, and a CSNN using the convolutional sizes of ResNet-18's top two layers (64 filters of 3×3 in the first layer and 128 filters of 3×3 in the second), the partitioned CSNN with a single layer showed about 5% higher accuracy with a single frame and over 10% improvement with 8 frames. These findings demonstrate that the proposed partitioned CSNN architecture, despite employing a relatively simple single-layer design, can achieve more efficient performance compared to conventional CNNs and more complex Spiking DCNN structures.

Finally, the computational efficiency of the partitioned structure is demonstrated by comparing the number of operations consumed by the entire network and the number of cycles required to read the sensor data. In the case of the 2-Layered CSNN presented in Table 1, because the convolution is performed after reading the entire sensor data, 32 cycles are required to read the TDM-based sensor data. By contrast, the partitioned CSNN processes the sensor data separately for each region, consuming 18 cycles based on reading the largest region, the palm area (18×18 pixels). This indicates a difference of approximately 60 ms in the time required to read the TDM-based sensor, which can be considered a substantial disparity in tactile systems that demand rapid response.

The computational load of the model can be measured using floating-point operations (FLOPs) required for the convolutional layers. Table 6 lists the computational loads required for the classification using a single frame. It can be confirmed that the proposed model has eight times less computational demand compared to the SDCNN used in previous research and exhibits 14.27 times less computational demand when compared to the CSNN with the same structure as the top two layers of ResNet-18. In other words, by utilizing a partitioned form of the CSNN, high-level interpretations can be achieved through the use of separate filters tailored to the size of each area, thereby ensuring high accuracy with minimal computational resources.

Table 6 Recognition accuracies, FLOPs and total number of required synapses of the proposed partitioned CSNN and 2-layered CSNN over the STAG dataset

Method	Readout cycles	FLOPs (M)	Total # of synapses
ResNet-18 (CNN) [38]	32	49.68	20,800
Spiking DCNN [39]	32	14.12	20,800
2-Layered (64-128) CSNN	32	24.84	74,880
Partitioned CSNN	18	1.74	9120

Furthermore, Table 6 quantifies the total number of synapses required for the entire network. This reveals that the partitioned network approach proposed in this study necessitates only 43% of the synapses used in previous research, while paradoxically achieving higher accuracy. Considering that prior studies have concluded that energy consumption from memory access is comparably as significant as that from the neural network's computational processes, the partitioned CSNN can be considered more energy-efficient than existing approaches. Synthesizing these results, it can be concluded that this approach is well-suited for the rapid computation and energy-efficient network configuration demanded by tactile perception.

7 Conclusion

In this study, a partitioned CSNN is designed for object recognition using tactile information. First, the input data from each hand region were preprocessed with Gabor filters and converted into a spike train with temporal information via intensity-to-latency encoding. These spikes are fed into partitioned SNNs, each consisting of a single convolutional and pooling layer, where feature information is extracted. The feature information extracted from each region is then combined into a single dataset through a concatenation layer and subsequently passed to a classifier for the final decision.

In the evaluation using the STAG dataset, the proposed model demonstrated superior performance while requiring fewer computations than the conventional unified CSNN structure. Notably, the model maintained higher performance in experiments involving multiple frames than existing methods. Although the proposed model performed well, a limitation was that the difference in accuracy between conventional CNNs increased as the number of frames increased when multiple frames were used. In future research, we plan to propose an approach that maintains computational efficiency without any degradation in performance compared to CNN-based methods in multi-frame environments. In addition, we aim to apply this model to large-scale tactile datasets to validate its suitability for tactile systems.

Author Contributions J.S.Go wrote the main manuscript text. All authors reviewed the manuscript.

Funding There was no funding for this work.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Liu W, Yu P, Gu C, Cheng X, Fu X (2017) Fingertip piezoelectric tactile sensor array for roughness encoding under varying scanning velocity. *IEEE Sens J* 17(21):6867–6879
2. Saal HP, Delhayé BP, Rayhaun BC, Bensmaia SJ (2017) Simulating tactile signals from the whole hand with millisecond precision. *Proc Natl Acad Sci USA* 114(28):5693–5702
3. Zhao L, Liu Y, Song W (2021) Tactile perceptual thresholds of electrovibration in VR. *IEEE Trans Visual Comput Graphics* 27(5):2618–2626
4. Zhang J et al (2024) Touchmark: partial tactile feedback design for upper limb rehabilitation in virtual reality. *IEEE Trans Vis Comput Graph* 30(11):7430–7440
5. Nagano R, Kinoshita T, Hattori S, Hiroi Y, Itoh Y, Hiraki T (2024) Haptofloater: visuo-haptic augmented reality by embedding imperceptible color vibration signals for tactile display control in a mid-air image. *IEEE Trans Vis Comput Graph* 30(11):7463–7472
6. Chun et al (2021) An artificial neural tactile sensing system. *Nat Electron* 4:429–438
7. Dahiya R (2019) E-skin: from humanoids to humans. *Proc IEEE* 107(2):247–252
8. Asghar W et al (2020) Piezocapacitive flexible E-skin pressure sensors having magnetically grown microstructures. *Adv Mater Technol* 5(2):1900934
9. Kang J et al (2018) Tough and water-insensitive self-healing elastomer for robust electronic skin. *Adv Mater* 30(13):1706846
10. Wang C, Huang Z, Xu S (2018) Materials and structures toward soft electronics. *Adv Mater* 30(50):1801368
11. Kim Y et al (2018) A bioinspired flexible organic artificial afferent nerve. *Science* 360(6392):998–1003
12. Khan WZ, Ahmed E, Hakak S, Yaqoob I, Ahmed A (2019) Edge computing: a survey. *Future Gener Comput Syst* 97:219–235
13. Navaraj WT et al (2017) Nanowire FET based neural element for robotic tactile sensing skin. *Front Neurosci*. <https://doi.org/10.3389/fnins.2017.00501>
14. Lin P, Chang S, Wang H, Huang Q, He J (2018) SpikeCD: a parameter-insensitive spiking neural network with clustering degeneracy strategy. *Neural Comput Appl* 31:3933–3945
15. Wang M et al (2021) Artificial skin perception. *Adv Mater* 33(19):2003014
16. Nunes JD, Carvalho M, Carneiro D, Cardoso JS (2022) Spiking neural networks: a survey. *IEEE Access* 10:60738–60764
17. Wang D et al (2023) A memristor-based learning engine for synaptic trace-based online learning. *IEEE Trans Biomed Circuits Syst* 17(5):1153–1165
18. Davidson S, Furber SB (2021) Comparison of artificial and spiking neural networks on digital hardware. *Front Neurosci*. <https://doi.org/10.3389/fnins.2021.651141>
19. Wijesinghe P, Ankit A, Sengupta A, Roy K (2018) An all-memristor deep spiking neural computing system: a step toward realizing the low-power stochastic brain. *IEEE Trans Emerg Top Comput Intell* 2(5):345–358
20. Zhao Z et al (2020) A memristor-based spiking neural network with high scalability and learning efficiency. *IEEE Trans Circuits Syst II Express Briefs* 67(5):931–935
21. Wu Z et al (2024) threshold switching memristor-based radial-based spiking neuron circuit for conversion-based spiking neural networks adversarial attack improvement. *IEEE Trans Circuits Syst II Express Briefs* 71(3):1446–1450

22. Wang D et al (2023) A memristor-based learning engine for synaptic trace-based online learning. *IEEE Trans Biomed Circuits Syst* 17(5):1153–1165
23. Kaboli M, Cheng G (2016) Novel tactile descriptors and a tactile transfer learning technique for active in-hand object recognition via texture properties. In: *Proc. IEEE-RAS Int. Conf. Humanoid Robots—Workshop Tactile Sens. Manipulation New Progress Challenges*
24. Kerr E, McGinnity TM, Coleman S (2018) Material recognition using tactile sensing. *Expert Syst Appl* 94:94–111
25. Gao J, Zhu X, Fu Z, Zhang W, Sun D, Gu W (2024) A grid-less flexible tactile sensing system based on deep neural network for two-point localization and shape recognition. *IEEE Sens J* 24(11):18259–18266
26. Li Z, Liu F, Yang W, Peng S, Zhou J (2022) A survey of convolutional neural networks: analysis, applications, and prospects. *IEEE Trans Neural Netw Learn Syst* 33(12):6999–7019
27. Zhu Q, Zu X (2022) Fully convolutional neural network structure and its loss function for image classification. *IEEE Access* 10:35541–35549
28. Tsuji S, Kohama T (2019) Using a convolutional neural network to construct a pen-type tactile sensor system for roughness recognition. *Sens Actuators A Phys* 291:7–12
29. Yuan W, Mo Y, Wang S, Adelson EH. (2018) Active clothing material perception using tactile sensing and deep learning. In: *2018 IEEE international conference on robotics and automation, Brisbane, Australia*, pp. 4842–4849
30. Kim et al., J (2020) Object shape recognition using tactile sensor arrays by a spiking neural network with unsupervised learning. In: *2020 IEEE international conference on systems, man, and cybernetics (SMC), Toronto, ON, Canada*, pp. 178–183
31. Parvizi-Fard A, Amiri M, Kumar D, Iskarous MM, Thakor NV (2021) A functional spiking neuronal network for tactile sensing pathway to process edge orientation. *Sci Rep*. <https://doi.org/10.1038/s41598-020-80132-4>
32. Gu F, Sng W, Taunayazov T, Soh H (2020) TactileSGNet: a spiking graph neural network for event-based tactile object recognition. In: *2020 IEEE/RSJ international conference on intelligent robots and systems (IROS), Las Vegas, NV, USA*, pp. 9876–9882
33. Peng K, Srutarshi B, Henry C, Aggelos K, Oliver C (2023) Boost event-driven tactile learning with location spiking neurons. *Front Neurosci*. <https://doi.org/10.3389/fnins.2023.1127537>
34. Xu L, Lin W, Kuo CCJ (2015) Visual quality assessment by machine learning, *Springer Briefs in Electrical and Computer Engineering*
35. Guo W, Fouda ME, Eltawil AM, Salama KN (2021) Neural coding in spiking neural networks: a comparative study for robust neuromorphic systems. *Front Neurosci*. <https://doi.org/10.3389/fnins.2021.638474>
36. Zhou S, Chen Y, Li X, Sanyal A (2020) Deep SCNN-based real-time object detection for self-driving vehicles using LiDAR temporal data. *IEEE Access* 8:76903–76912
37. Mozafari M, Ganjtabesh M, Nowzari-Dalini A, Masquelier T (2019) SpykeTorch: efficient simulation of convolutional spiking neural networks with at most one spike per neuron. *Front Neurosci*. <https://doi.org/10.3389/fnins.2019.00625>
38. Sundaram S, Kellnhofer P, Li Y, Zhu J, Torralba A, Matusik W (2019) Learning the signatures of the human grasp using a scalable tactile glove. *Nature* 569:698–702
39. KJheradpisheh SR, Ganjtabesh M, Thorpe SJ, Masquelier T (2018) STDP-based spiking deep convolutional neural networks for object recognition. *Neural Netw* 99:56–67